

Lab 4

1. Write a program that takes two integers from the user: total_runs and innings_played. Calculate and display the player's batting average.
2. Write a program that takes an integer in decimal format and print it in octal, hexadecimal formats
3. Write a program that takes an integer in octal or hexadecimal format and print it in decimal format. Hint: use %i
4. Assume that a class has 5 students. Read the Roll number (integer) and Marks out of (100) for the 5 students. Compute the average mark (float) for each student and display the data nicely formatted.

Example:

Enter details for student 1:

Roll: 1001

Marks: 75 35 95 50 99

Enter details for student 2:

Roll: 2435

Marks: 50 45 99 80 83

Enter details for student 3:

Roll: 1023

Marks: 75 55 89 77 56

Enter details for student 4:

Roll: 2132

Marks: 88 77 98 78 64

Enter details for student 5:

Roll: 3152

Marks: 54 20 21 86 55

Tabular Display:

Roll: 1001 Marks: 75 35 95 50

99 Average= 70.8

Roll: 2435 Marks: 50 45 99 80

83 Average= 71.4

Roll: 1023 Marks: 75 55 89 77

56 Average= 70.4

Roll: 2132 Marks: 88 77 98 78 64

Average= 81

Roll: 3152 Marks: 54 20 21 86 55 Average= 47.2

5. Read three pairs of floating-point integer values (x1, y1) , (x2, y2) , and (x3, y3) representing three points on a plane. Determine if they are collinear. Display your result.

6. Write a program that takes an integer between 1 and 9 as input, and prints out on the terminal a pattern similar to the following. The example display shown is for input 5.
Example (n=5):

```
1           1
2 2        2 2
3 3 3      3 3 3
4 4 4 4    4 4 4 4
5 5 5 5 5 5 5 5 5 5
```

7. Write a C program that asks the user to enter a single lowercase alphabet character (e.g., 'a') and an integer cipher key (e.g., 3). Print its Capital form and shift it forward by the key using ASCII values (e.g. A Cipher of 'a' is 'd').
8. Write a program that prompts the user to enter a floating-point number (e.g., 12.345). The program must isolate and print the whole number part and the fractional part separately. (**Input:** 45.678, **Output:** Whole part: 45, Fractional part: 0.678)
9. Write a program that simulates a website tracking unique visitor views. Create a function called void visitPage() that prints how many times it has been called. Note: You are **not allowed** to use global variables. You must use a local variable inside visitPage() that retains its value across multiple function calls.
10. Write a program that processes a massive loop like iterating a few hundred million times to sum up numbers. Use a register variable.
11. Write a program to declare a const variable and show that its value cannot be changed.
12. Write a program to check whether a given character is uppercase or lowercase (without using any predefined function).
13. Write a program to declare variables using short, long, unsigned int and display their sizes and values.
14. Write a program to print the size (in bytes) of all basic data types using sizeof().